DMG / CYD Game Boy - Wiring Reference

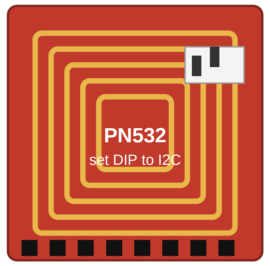
ESP32-2432S024 (E32R24P) | 2.4in ST7789 240x320 | board shown from the BACK | single I2C bus, nothing soldered to the CYD | rev B

SPEAKER 2-pin JST

8 ohm, 0.5-1 W, 36 mm.
Reuse the original DMG driver.
IO4 = amp enable (HIGH = on).
Mute it when the APU is idle.
Remap LED_R_PIN off IO4.

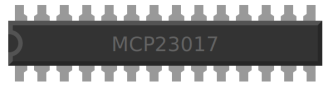
PN532 NFC READER (I2C mode)

Set the DIP switches to I2C.
Address 0x24 - clear of 0x20.
VCC 3.3V GND SDA SCL
Read once at boot / cart insert.
UID alone is enough to pick a ROM.

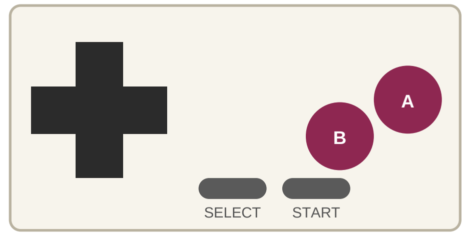


MCP23017 I2C GPIO EXPANDER

A0/A1/A2 -> GND (addr 0x20)
RESET -> 3.3V (never float it)
VDD 3.3V VSS GND SDA SCL
GPPU = 0xFF (internal pull-ups)
INT unused - poll once per frame.



BUTTON PCB

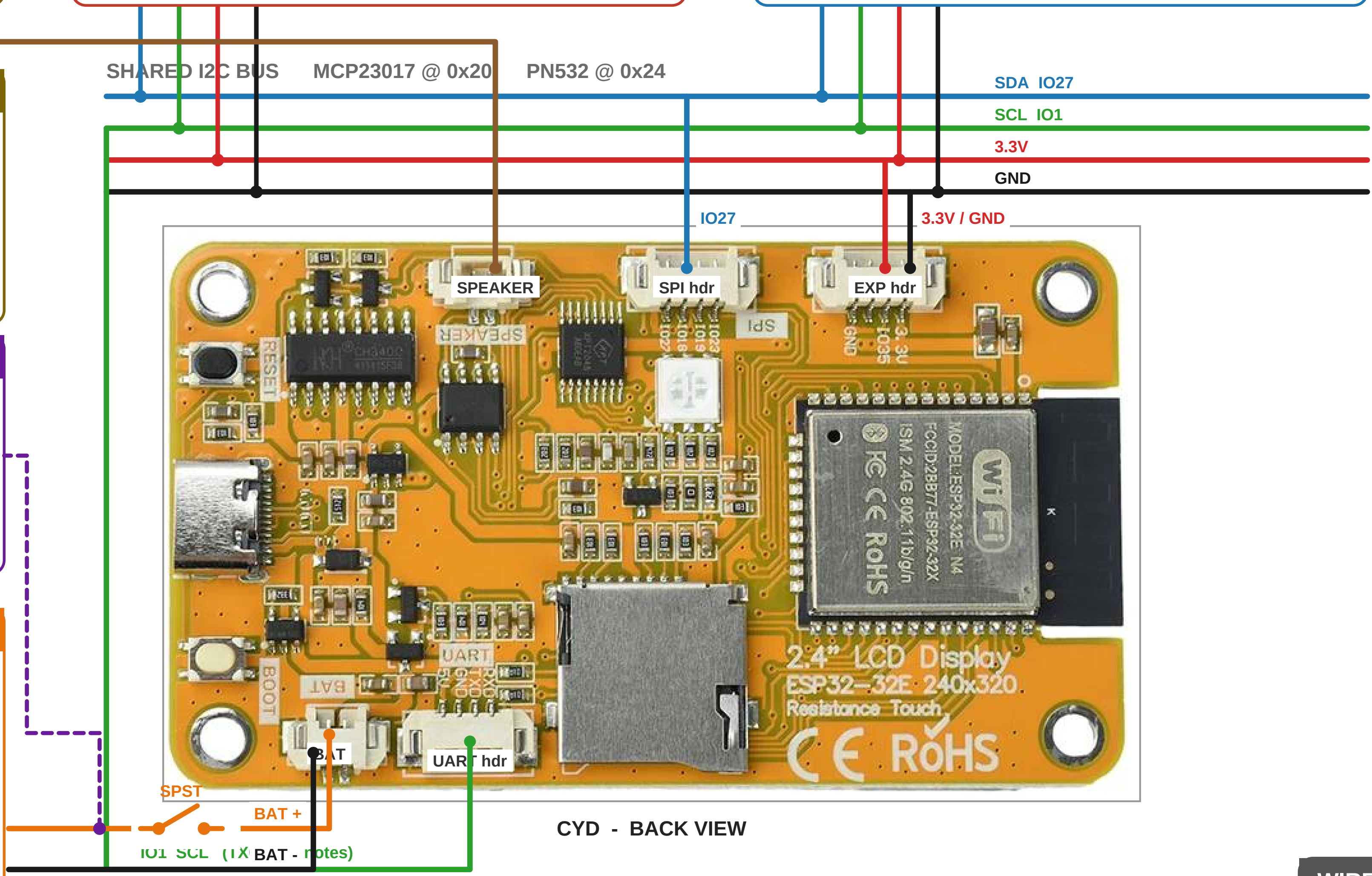


8 outputs + common GND

GPA0 Up
GPA1 Down
GPA2 Left
GPA3 Right
GPA4 A
GPA5 B
GPA6 Start
GPA7 Select
GPB0 Menu (spare)

Common GND -> MCP VSS.
Active LOW.

Debounce approx 8 ms in software; the expander has no hardware filter.



CYD - BACK VIEW

CARTRIDGES

NTAG215 disc, 25 mm, inside an empty DMG cartridge shell.
5 game carts locked; 1 wildcard left writable. Write with a phone, never on-device.
Remove the DMG RF shield or the read will fail.

OPTIONAL I2S AMP POWER

VIN -> switched BAT + GND -> BAT -

NOT the 3.3V rail - the amp draws 200-250 mA in bursts. 2.5-5.5 V input, so cell voltage gives ~1 W into 8 ohm.
Put the 100-220 uF cap AT the amp.

POWER onboard charger + switch

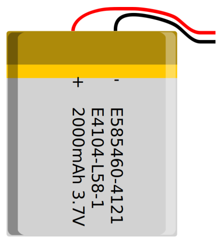
LiPo (+) -> SWITCH -> CYD BAT +
LiPo (-) -------------> CYD BAT -

3.7 V pouch, ~2000 mAh, WITH a protection PCB. Max approx 60x35x7 mm.

CHARGE WITH THE SWITCH ON.
The switch cuts the cell off from the whole board, charger included. Switch off means no charging.

Charging uses the CYD's own onboard charger. Run a USB-C pigtail into the battery bay - the door pops off with no tools.

Meter polarity before first connect.
JST is not a polarity standard.



BOOT ORDER & BUS NOTES

SCL must be IO1, not SDA. IO1 is TX0: the ROM bootloader prints on it at every power-up.
With SCL there, that traffic is only clock pulses with no START, so every device ignores it.
Reversed, you would generate a burst of spurious STARTs and STOPs on the bus.

Never put I2C on IO3. IO3 is RX0, driven push-pull by the USB-serial chip whenever USB is powered - including while charging. Open-drain I2C cannot pull against it.

Run bus recovery before Wire.begin(27, 1): nine manual SCL pulses, then a STOP. 400 kHz is fine.

If the 4th EXP-header pad meters out as IO22, use SCL = IO22 and none of the above applies.
microSD stays onboard on IO5/18/19/23 and is no longer shared with the reader.

WIRE COLOUR KEY

blue	SDA IO27 (SPI hdr)
green	SCL IO1 (UART hdr)
red	3.3 V (EXP hdr)
black	GND (EXP hdr)
orange	Battery + via switch
brown	Speaker +/-
dark grey	Buttons GPA0-7

METER EVERY HEADER PIN FIRST
EXP silkscreen reads GND / IO35 / 3.3V; the 4th pad is unlabelled.

Component artwork for the expander, cell and speaker from the Fritzing parts library (CC BY-SA 3.0). Board photo from the vendor datasheet. Schematic reference - not to scale.

Parts List, Pin Map & Build Notes

BILL OF MATERIALS

CORE		
1x	ESP32-2432S024 / E32R24P	2.4in ST7789 CYD, USB-C, LiPo charger
1x	microSD card 8-32 GB	FAT32
I2C BUS (both devices share 4 wires)		
1x	MCP23017 breakout	16 GPIO, addr 0x20
1x	PN532 NFC module (red board)	DIP-selectable I2C - NOT the RC522
2x	4.7k resistor	SDA/SCL pull-ups if neither breakout has them
INPUT		
1x	Button PCB, 8 out + GND	your existing board
1x	DMG membrane + rubber pad set	D-pad, A/B, Start/Select
NFC CARTRIDGES		
Nx	NTAG215 disc, 25 mm	one per cartridge
Nx	Empty DMG cartridge shell	houses the disc
POWER + AUDIO		
1x	TP4056 module WITH protection	6 pads: IN+/- B+/- OUT+/-. Needs DW01A + FS8205.
1x	3.7V LiPo pouch ~2000 mAh	approx 60 x 35 x 7 mm (AA bay)
2x	5.1k resistor	only if the TP4056 lacks CC1/CC2 pull-downs
1x	2.4k resistor	Rprog for 500 mA, if the board ships at 1.2k
1x	8 ohm 0.5-1 W speaker, 36 mm	reuse the DMG original
1x	100-220 uF electrolytic	mount AT the amp, not near the board
MECHANICAL + CONSUMABLES		
1x	DMG-01 shell + screen lens	cutout measured 47 x 42.5 mm
1x	3D-printed bezel	aperture 36.0 x 32.4 mm, 1.5 mm rear flange
2x	JST 1.25mm 4-pin pigtail	SPI hdr, EXP hdr (one ships in the box)
1x	JST 1.25mm 2-pin pigtail	SPEAKER (BAT lead ships in the box)
1x	JST 1.25mm 4-pin pigtail	UART hdr - for SCL
-	Buy pre-crimped pigtails	1.25 mm hand-crimping is miserable
-	30 AWG silicone wire, Kapton, foam tape, M2 inserts	

PIN MAP

SIGNAL		
I2C SDA	IO27	SPI header pin 1
I2C SCL	IO1	UART header (TX0) - see boot note
3.3V / GND	-	EXP header
Amp enable	IO4	HIGH = on. Mute when APU idle.
Audio out	IO26	DAC to onboard amp
Battery sense	IO34	ADC
SD CS	IO5	onboard
SD SPI	IO18/19/23	onboard - now exclusive to SD
Backlight	IO21	PWM brightness
LCD	IO15/2/14/13	CS / DC / SCK / MOSI
TFT_RST	-1	panel reset ties to EN
Unused	IO35, IO22?	IO35 input-only; meter the 4th EXP pad

FIRMWARE NOTES

1. Scaling 24/16 (3/2) -> 240 x 216 px = 36.0 x 32.4 mm. Bezel border approx 5.1 mm sides, 5.0 mm top and bottom.
2. fillScreen(TFT_BLACK) once at init, then only ever write the 240 x 216 window.
3. User_Setup: ST7789_DRIVER, TFT_RST -1, try SPI_FREQUENCY 80000000.
4. Enable PEANUT_GB_12_COLOUR; use a flat 64-entry LUT indexed by the raw pixel byte (values 0x00-0x23).
5. Blend the interpolated pixel: a, (a+b)/2, b - both axes. Blend AFTER the LUT so no palette bits ride along.
6. Split cores: Peanut-GB on core 1, display push on core 0 behind a queue, or pushPixelsDMA with two row buffers.
7. Battery saves only. peanut-gb has no save-state API. Use gb_get_save_size_s() and dump the cart RAM buffer.
8. Flush the .sav on pause and on a low-battery threshold read from IO34 - there is no cart-removal event.
9. Map NFC tag UID -> ROM filename in carts.txt on the SD card. Reading the UID alone is enough.
10. Remap LED_R_PIN. The stock hw_config.h sets it to 4, which is the audio amp enable on this board.
11. Test speaker gain with tone() on IO26 before modifying any resistors.

POWER NOTES

A. CHARGE WITH THE SWITCH ON. The switch cuts the cell from the whole board, onboard charger included. Switch off = no charging.
B. Switch OFF with USB plugged in still runs the board off the regulator, with the battery isolated. Detect it and say 'switch on to charge'.
C. Detect charging by trend, not threshold: sample IO34 every 5 s; rising over 60 s means charge is going in. A resting and a held cell look identical.
D. Charging screen must be low power - 80 MHz CPU, backlight ~10%, IO4 low, emulation paused. Otherwise the load eats most of the 500 mA charg
E. Add a soft cutoff on IO34: ~3.5 V flush the .sav and warn, ~3.3 V flush and deep sleep.
F. Meter cell polarity against the BAT silkscreen before first connection. JST is not a polarity standard.
G. Optional I2S amp takes VIN from the switched battery line, NOT the 3.3 V rail. Ground to battery negative on the same harness.
12. PCB short axis is 42.89 mm against a 42.5 mm cutout - about 0.2 mm margin. Use a rear flange on the bezel.

Pin assignments for IO27 / IO1 are proposals derived from the vendor pin table and board photo. Confirm continuity with a meter before connecting anything.